Asu Singh

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PROFESSIONAL SUMMARY

Software Development Engineer with 5+ years building production financial-services software, approaching six years, and hands-on experience using and building agentic AI systems for personal projects. Strong foundation in Java, Python, event-driven systems, performance engineering, database reliability, observability, and machine learning. Published peer-reviewed research in Cognitive Computation on a Hybrid Social Group Optimization method for COVID-19 chest X-ray detection, achieving over 99% accuracy. Interested in applying production engineering, empirical evaluation, and practical AI-system building to reliable, beneficial AI deployments.

CORE COMPETENCIES

Applied AI Engineering | Agent Architectures | Empirical Evaluation | Production Reliability | Event-Driven Systems | Observability & Debugging | Java / Python | Technical Communication

WORK EXPERIENCE

ION Group

Aug 2020 - Present

Software Development Engineer

Noida, India

- Designed and developed XTP, a real-time back-office trading product used in financial-services operations.
- Worked with an in-house MQ-like event-driven notification system, understanding its internal workings and operational failure points.
- Led Java multithreading refactoring that increased component performance by 20x.
- Developed Alazium, a log-analysis initiative that reduced production issue-analysis time by 30%.
- Designed a recovery feature for halted end-of-day processing, reducing downtime by 90%.
- Redesigned historical-data cleanup using a tweaked linear-regression approach, improving database performance and reducing processing time by 95% under load.
- Led migration of XTP CI systems from Red Hat 7 to Red Hat 8, supporting reliable engineering delivery.

Holisol Logistics

Jun 2019 - Sep 2019

Software Development Intern

Delhi, India

- Developed a Daily Run Sheet Generator with a Django frontend to optimise deliveries and presented the application to the company founder for deployment approval.

SELECTED PROJECTS & PUBLICATION

Agentic AI Personal Projects — Applied AI

Used agentic AI and created an agentic AI system for personal projects; practical experience with agent workflows and the evolving production considerations around reliable AI systems.

Agentic AI

Hybrid Social Group Optimization for COVID-19 Detection — Publication

Developed a meta-heuristic method for COVID-19 detection from chest X-ray images, achieving over 99% accuracy. Published in the peer-reviewed Cognitive Computation journal.

Machine Learning, Computer Vision

Lung Disease Prediction — Machine Learning

Developed an automated application to detect and predict the onset of lung diseases from X-ray images, achieving over 99% accuracy.

Machine Learning, Medical Imaging

EDUCATION

Bachelor of Technology Maharaja Agrasen Institute of Technology, Delhi 2016 - 2020

CERTIFICATIONS

TECHNICAL SKILLS

Languages: Java, Python, C++, C#

Backend & Data: Java 8+, Spring Boot, Microservices, REST APIs, Hibernate, Oracle, PostgreSQL, Django

Engineering: Multithreading, Performance Tuning, Event-Driven Systems, MQ-like Messaging, TDD, JUnit, Mockito, Unix

Cloud & Observability: AWS, Grafana, Prometheus, Production Debugging, Log Analysis

AI & ML: Agentic AI, Machine Learning, Computer Vision, Empirical Evaluation